

Deleted from the Router, Not from the Model: What Selector-Based Unlearning Actually Removes

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Abstract

A growing family of “exact unlearning” systems makes deletion cheap by making it modular: train one small expert per data source, keep a selector that routes each query to an expert, and delete a source by dropping its expert. The deletion is exact in the only sense the architecture promises — no surviving parameter was ever trained on the removed data. We show that this promise is compatible with a system that still answers questions about the deleted person, in detail, using a stranger’s biography, and that the field’s standard forget metric rates that behaviour as successful unlearning.

We audit deletion-under-a-selector as a design pattern on a 200-expert LoRA pool over TOFU, deleting 20 of 200 sources. Four results. (i) *The benchmark cannot see substitution*: a “method” that deletes nothing and merely reroutes the deleted sources’ queries to one arbitrary surviving expert scores at or above genuine deletion on TOFU **forget_quality** in six of seven destination choices at byte-identical utility; the metric moves 0.53 on a choice that has nothing to do with forgetting. (ii) *The substitution is substantive*: orphaned queries are answered with a specific surviving author’s distinctive facts at a rate of 0.24–0.30 against a random-destination floor of 0.17, and refusal essentially never happens (≤ 0.013 over 1600 answers). (iii) *Every reassuring property of the pattern is measured on queries that name the person being deleted*: orphan detectability, deletion locality, and routing accuracy all collapse when the name is removed, in every selector family but one. (iv) *The one exception does not yield a deployable defense*: a perplexity-based refusal gate — the only selector that survives anonymisation, and which we make 45–90 $\times$ cheaper at no loss — is perfect on name-bearing queries and refuses 41.8% of legitimate traffic to catch 90% of orphans on anonymised ones. It works exactly where it is not needed.

We argue that a routed system’s deletion claim is not meaningful without reporting what it substitutes, and we release the metric suite (ORR, CSAR, RDR) and the audit harness used here.

1 Introduction

Machine unlearning asks a system to behave as though it had never seen some training data [Bourtoule et al., 2021]. Because retraining a large language model from scratch is prohibitive, a substantial line of work makes deletion cheap *architecturally*: partition the training data by source, train an isolated module per source, and compose the modules at inference [Yu et al., 2022, Basu Roy Chowdhury et al., 2025, Zhao et al., 2024, Grimes et al., 2026, Schneider et al., 2026]. Deleting a source is then dropping its module. This is attractive precisely because the guarantee is structural rather than statistical: after the drop, no surviving parameter was ever updated on the removed data.

That guarantee is real, and it is narrower than it sounds. A modular system needs a *selector* — a router, retriever, gate, or key matcher — to decide which module answers a query. Deletion removes a module; it does not remove the queries that used to be routed to it. Every one of those

queries still arrives, and the selector must send it somewhere. We call such a query an *orphan*, and the question this paper asks is simple: **what does the system say to an orphan?**

The answer, on a per-source expert pool over TOFU [Maini et al., 2024], is that it answers confidently with someone else’s life. Asked whether a deleted author had won any awards, the system names an award — one that belongs to a different, still-present author, with the name substituted and the rest of the sentence intact. It does this on roughly a third of orphaned queries, and it refuses on approximately none of them. The deletion was exact. The output is a fabricated biography attached to a real name, and the person the facts actually belong to did not consent to that either.

Worse, the field’s standard instrument reports this as success. TOFU’s `forget_quality` is a two-sample statistical test comparing the post-deletion model’s behaviour on the forget set against a reference model retrained without it. It is a test of *distributional indistinguishability*, and a stranger’s answers are, distributionally, about as far from the deleted author’s memorised answers as a retrained model’s are. We exhibit an “unlearning method” that deletes nothing — it only redirects — and scores at or above genuine deletion.

Contributions.

1. **A deletion semantics for selector-based systems (§2.2)**, separating what the architecture guarantees (D1: no surviving parameter saw the data) from what a data subject would expect (D2: the system does not answer as though it still held the source; D3: an adversary cannot tell that a deletion occurred).
2. **A metric suite for what deletion substitutes (§3)**: orphan response rate (ORR), cross-source attribution rate (CSAR), and retained displacement rate (RDR), together with a query-transform protocol that separates a property of the system from a property of the benchmark’s phrasing.
3. **Evidence that TOFU’s forget metric is blind to substitution (§5)**, with paired confidence intervals and an explicit account of the metric’s resolution.
4. **A measurement of the harm (§6)** at fact level rather than by n -gram overlap, decomposed into substantive attribution and bare name substitution.
5. **A defense frontier that ends negatively (§9)**: the one record-free detector that survives anonymisation is cheap but not deployable, and we report where its false-refusal cost binds.
6. **The audit harness and the negative results**, including four hypotheses we pre-registered and then refuted, and eight measurement defects that produced plausible numbers before they were caught (§11).

2 Background and Threat Model

2.1 The pattern under audit

We study systems of the following shape, which covers keyed adapter banks [Yu et al., 2022], sharded PEFT unlearning [Basu Roy Chowdhury et al., 2025], retrieval-routed LoRA pools [Zhao et al., 2024], per-user deletable proxies [Schneider et al., 2026], and block-listed memory modules [Grimes et al., 2026]:

A frozen base model f_0 ; a set of k modules $\{\phi_1, \dots, \phi_k\}$, module i trained only on source i ; and a selector S mapping a query q to one module (or a small set). The served answer is f_0 composed with $\phi_{S(q)}$. Deleting source i means removing ϕ_i from the pool and from S 's candidate set.

The selector is the part that has not been audited. It is usually treated as an engineering detail — an embedding nearest-neighbour, a TF-IDF key match, a learned gate — and its behaviour *after* a deletion is not part of any published deletion claim we are aware of.

2.2 Deletion semantics: D1, D2, D3

We distinguish three properties that “the source was deleted” can mean.

D1 (parametric). No parameter serving the query was trained on the deleted source. This is what modular architectures guarantee, and they guarantee it by construction.

D2 (behavioural). The system does not answer as though it still held the source. This is what a data subject, and most readers of an unlearning paper, take “deleted” to mean. D1 does not imply D2: a substituted expert answers fluently and specifically, just wrongly.

D3 (indistinguishability). An observer with query access cannot determine that a deletion occurred. D1 does not imply D3 either — deletion changes the selector’s score geometry, and that change is measurable from the surviving units alone.

The gap between D1 and D2 is the subject of §5–§6; the gap between D1 and D3 is the subject of §8–§9. A system may satisfy D1 while violating both.

2.3 Threat model

We consider two parties, and deliberately give them *less* than the deployment usually would.

The adversary has query access only. No deletion record, no list of which sources exist, no model weights, no training data. They observe the served answer and, in the strongest variant we allow, the selector’s own scores over the *surviving* modules — which a system that exposes retrieval confidence or expert attributions already hands out. Their goals are D3 (detect that a deletion happened) and steering (make a chosen module answer a chosen query).

The defender is likewise record-free: they may not consult a list of deleted sources, because the interesting question is whether the system can refuse an orphan *without* maintaining exactly the registry of deleted people that deletion was supposed to eliminate. A defender permitted a per-source tombstone is not solving a hard problem; on our pool that trivially reaches AUC 0.982, and it is a record of the deletion, which is the thing under dispute.

3 Metrics

TOFU supplies `forget_quality` and `model_utility`; neither is a statement about what the system substitutes. We add three.

ORR — orphan response rate. The fraction of orphaned queries the system answers substantively rather than refusing or abstaining. Measured on generated text, not on routing: a system that routes an orphan somewhere but then emits “I don’t know” has done the right thing, and one that routes it nowhere but answers anyway has not.

Transform	Construction	Role
gold	TOFU’s question verbatim (names the author in $\sim 90\%$ of rows)	anchor
paraphrase	TOFU’s own paraphrase	null by construction — keeps the name
name_stripped	all name forms removed	the hard case
indirect	name replaced by a description built from the author’s own facts	weaker than it looks (§9)
para_stripped	TOFU’s paraphrase, name removed	the independent hard surface

Table 1: The query-transform axis. **paraphrase** is included because it is the obvious robustness check and it reports a reassuring nothing: it preserves the author’s name (coverage 0.900 vs. 0.895 for the gold form), so it perturbs everything except the feature the selectors actually use.

CSAR — cross-source attribution rate. The fraction of orphan answers that assert at least one fact that is (i) *distinctive* to the surviving source the query was routed to (appears in the gold answers of at most two of the 200 sources), (ii) *not* the deleted source’s own fact — serving the deleted source’s real content is a disclosure of the deleted source, a different and worse failure that must not be folded in — and (iii) *not* present in the frozen base model’s answer to the same question, so the base’s own knowledge is not credited to the selector.

CSAR is deliberately not an n -gram metric. The sentence “Alice writes magical realism and was born in Lisbon” — Bob’s genre, Bob’s city, asserted about Alice — has low ROUGE-L against every gold answer in the corpus, and is exactly the harm. On our pool a ROUGE-based audit files 95.5% of orphan answers as novel confabulation; CSAR finds a named stranger’s facts in roughly half of that “no leak here” bucket.

We further decompose CSAR into a *substantive* part (at least one matched fact is not merely a name-form of the routed survivor) and a *name-only* part. Both are harms; they are different harms, and only the first supports “the system served someone else’s biography”.

RDR — retained displacement rate. The fraction of *retained* queries that change module as a side effect of a deletion nobody asked for. Deletion is supposed to be local to the deleted source; RDR measures whether it is.

Detection AUC. Whether an orphan query can be told from a retained one using only the surviving selector’s scores, with no deletion record consulted. This operationalises D3, and, read from the defender’s side, it is the ceiling on any record-free refusal gate.

3.1 Query transforms

A harm measured only on queries that name the deleted person is worth as much as a defense measured that way. Every result below is reported across a transform axis (Table 1); the perturbation is applied to *both* classes, so no detector can learn “perturbed $\Rightarrow$ deleted”.

4 Experimental Setup

Pool. Llama-2-7B-chat as the frozen base, with $k = 200$ LoRA experts [Hu et al., 2022], one per TOFU author, rank 32, 25 epochs. TOFU’s 200 fictitious authors with 20 QA pairs each give a

Destination	s89	s137	s31	s97	DELETE	s33	s79	s88
expert affinity	.3044	.3382	.2840	.2267	—	.2663	.2193	.3970
forget_quality	.8958	.8002	.6288	.6288	.5140	.5140	.5140	.3615

Table 2: TOFU `forget_quality` across reroute destinations. `model_utility` is *identical* to 0.8009 in all eight arms, and route audits confirm `deleted: 0 / rerouted: 1320` in every reroute arm against `deleted: 1320 / rerouted: 0` in the baseline. Six of the seven arms that delete nothing score at or above genuine deletion.

clean ground truth for who knows what. Deletion is TOFU `forget10`: authors 180–199, i.e. 20 of 200 sources and 400 orphaned questions. Two further pools (rank 8 / 5 epochs, rank 32 / 5 epochs) and a third built for §10 (rank 32 / 50 epochs) separate recipe effects from the pattern.

Selectors. Eight, in three families. *Lexical*: `key_exact` (author-name substring match), `key_tfidf`. *Feature-space*: `centroid_sbert` (MiniLM question embeddings [Reimers and Gurevych, 2019]), `centroid_lm` (base-model embeddings with adapters disabled). *Behavioural*: `ppl` (per-expert loss on the query), `activation_norm`, `attn_norm`, `logit_div`. A property worth stating once: *no feature-space selector reads any expert’s weights*, which we verified by finding their score matrices byte-identical across pools that differ in rank and epochs. Recipe questions can therefore only be asked of the behavioural family — a control over a variable the measurement does not consume is not a control.

Evaluation protocol. All detection numbers use an author-parity split: even-numbered authors fit, odd-numbered authors evaluate, so every evaluated deleted source is one the detector never saw. Features are permutation-invariant functions of the sorted survivor scores, never raw column identities, so a detector fit on one deletion transfers to an unseen one. Every serving arm audits the *authors* taking each route before any metric is read — a plausible-but-wrong route is the failure mode these arms are most exposed to, and no downstream metric would flag it.

5 The Forget Metric Is Blind to Substitution

Construction. We define E5, an “unlearning method” that deletes nothing. All 400 orphaned queries are rerouted to one fixed surviving expert; every expert, including the 20 supposedly forgotten ones, remains in the pool and in memory. We score seven destination choices and one genuine-deletion baseline (experts dropped, orphans served by the frozen base) at the same tier, against the same held-fixed KS reference, so all eight cells are the same statistical test.

Result. Table 2. Six of seven arms that delete nothing score at or above genuine deletion; two land exactly on it. `forget_quality` spans 0.3615–0.8958 — a swing of 0.53 — on a choice that holds fixed whether anything was forgotten, while `model_utility` does not move at all.

The obvious explanation, that rerouting to a similar author makes the answers look right, is *refuted*: Spearman correlation between `forget_quality` and destination affinity is +0.109 ($p = 0.82$), and the sign is nominally opposite to what the hypothesis predicts. It fails at a coarser tier too ($\rho = -0.059$, $p = 0.88$).

Condition	CSAR (centroid_sbert)	CSAR (key_tfidf)	own-disclosure	refusal
gold	0.3325	0.3650	0.913 / 0.933	0.000
name_stripped	0.4400	0.4175	0.273	≤ 0.010
indirect	0.3350	0.2125	0.383 / 0.393	≤ 0.013
random destination	0.2200	—	0.953	0.003

Table 3: Cross-source attribution over 400 orphan queries per router. “own-disclosure” is the rate at which the served answer contains the *deleted* author’s own distinctive facts — a different failure, reported separately so it is not folded into CSAR.

Is the spread real? The arms score identical rows, so this is a paired design and requires a paired interval; bootstrapping each arm’s marginal re-adds the noise the arms hold in *common*, once per arm, and can never resolve any spread (§11). Resampling row indices once per iteration and applying that draw to every arm gives a spread 95% CI of [0.2245, 0.6975] with $P(\text{spread} > 0.25) = 0.961$, and an at-or-above-deletion count of 6/7 observed with 95% CI [2, 7]. Even at the pessimistic end, two arms that delete nothing match or beat real deletion.

Two limits that travel with these numbers. First, the *ordering* does not reproduce across evaluation tiers (Spearman extended~smoke = +0.620, $p = 0.14$): the spread replicates, the per-destination ranking does not, so no claim may name a winning destination. Second, a single `forget_quality` cell is worth about ± 0.35 (marginal 95% CI widths 0.63–0.72). The KS statistic moves on an exact lattice of $1/\text{lcm}(n, m)$ — here $1/120$, one forget question — with roughly 30 attainable p -values above 0.05. The four decimals this literature reports are spurious precision.

Reading. `forget_quality` asks whether the forget-set behaviour is distributionally distinguishable from a retrained reference. A stranger’s answers satisfy that as well as an absence does — better, on most destinations. The metric was never designed to detect substitution, and it does not; but it is being used as the field’s evidence that deletion worked.

6 What the System Actually Says

Table 3 gives three results.

Refusal never happens. ≤ 0.013 across 1600 answers: $ORR \approx 1.00$ at the level of what is *said*, not merely where the query is routed. The information needed to refuse is often present — `key_exact` raises no-match on 100% of orphans, then falls back to `candidates[0]` and answers anyway — and is discarded.

Anonymising the query does not reduce the harm; it changes its character. Strip the name and the system leaks *less* of the deleted author’s own content ($0.913 \rightarrow 0.273$) while attributing *more* of a stranger’s content to them ($0.3325 \rightarrow 0.4400$). Attribution does not require the router to be *right*, only confident enough to pick someone: whichever expert is activated asserts its own facts, and worse routing produces *more legible* attribution, because an unrelated survivor’s facts register cleanly while a similar survivor’s overlap the deleted author’s and are excluded by CSAR’s own-facts filter.

It is not a naming glitch. The strongest objection to CSAR is that it may be counting the router emitting the survivor’s *name* in place of the deleted author’s. Splitting every cross-source row by whether its matched facts are exhausted by name-forms of the routed survivor:

Router	CSAR	substantive	name-only
centroid_sbert	0.3325	0.2400	0.0925
key_tfidf	0.3650	0.2950	0.0700
random floor	0.2200	0.1725	0.0475

Two thirds to four fifths of CSAR carries a real fact — a title, a place, an award, an occupation. It survives dropping the identity-question slice, where a wrong expert answers with a name by construction: on the 300 non-identity queries substantive CSAR is 0.217/0.250, still above the floor. **We therefore quote the substantive figure (0.24–0.30 against a 0.17 floor), not the raw 0.33–0.37**, with name-only as its own row.

The floor matters. A uniformly random surviving expert — no router at all — already produces $CSAR = 0.2200$ (0.1725 substantive). The harm therefore *cannot be engineered to zero by improving the selector*; roughly two thirds of it is a property of substituting any expert. It also means a pre-registered “CSAR > 0.20” bar is nearly uninformative on its own, and the real quantity is the lift over the random-destination floor.

[BLOCKED: The pre-registration requires ~300 hand labels validating the CSAR classifier before any CSAR number is published. Records are staged; the classifier’s author cannot validate it. No CSAR number in this section may go to a venue until those labels exist.]

7 Where Orphans Go, and What Deletion Disturbs

Orphans are all reassigned. 400/400 deleted-author questions are answered by some surviving expert under every selector. At $k = 200$ the mass is diffuse — busiest survivor 0.11–0.19, effective number of destinations 17.4–24.2 — except for **key_exact**, which sends 100% of them to one unit, because it returns the first shard *by index* whose name appears and falls back to **candidates[0]** otherwise.

Magnet saturation is refuted. A natural prediction is that deleting sources one at a time makes one survivor progressively the answerer for a growing share of the corpus. It does the opposite in six of seven cells: with one author deleted, all 20 of their questions go to that author’s single nearest survivor (share 0.55–0.75); with twenty deleted, each has a *different* nearest survivor, so the pooled share falls and the effective count rises to 17–29. The prediction holds only where the router cannot tell the deleted sources apart — and there it holds sharply: **key_tfidf** on **indirect** queries saturates at 0.902 with an effective count of 1.2, one survivor answering nine of every ten orphans.

That survivor is a structural artifact. It is one of **18 of 200 authors from whom no name can be extracted**. Its centroid is the most generic point in the space; it already absorbs 68% of real-author and 45% of world-facts out-of-distribution queries. The routing magnet and the missing-name artifact are the *same* units, so any survivor-conditioned statistic is least trustworthy

in exactly the name-free conditions this paper most wants to report. We flag every such cell rather than dropping it (§11).

Deletion locality is lexical too. RDR is 0.000 on gold-form queries at $k = 200$ (against 5.8% published at $k = 10$, so fine-grained units genuinely do help locality) — but 0.092 name-stripped for `centroid_sbert`. Deleting 20 of 200 sources moves 9.2% of *retained* queries to a different expert, worse than the coarse-unit number it was supposed to improve on. Locality, like detectability, is a property of the query naming its source.

8 Detectability Is a Lexical Artifact, With One Exception

Granularity, not confidence, is the axis. The published finding that confidence-based orphan refusal caps at AUC 0.57–0.61 turns out to be a statement about *coarse units*, not about selectors. At constant deletion size, detection rises monotonically with k : `centroid_sbert` 0.564 $\rightarrow$ 0.795 $\rightarrow$ 0.984 and `centroid_lm` 0.502 $\rightarrow$ 0.628 $\rightarrow$ 0.761 for $k = 10, 50, 200$, with `key_tfidf` saturated throughout (0.973 $\rightarrow$ 0.989). Attribution recall — naming *which* deleted source an orphan belongs to — rises 0.300 $\rightarrow$ 0.700 $\rightarrow$ 1.000.

And it is a lexical artifact. Strip the name and the ladder flattens to 0.560 $\rightarrow$ 0.572 $\rightarrow$ 0.623: +0.063 across a $20\times$ granularity change, against +0.367 on gold-form queries. `key_tfidf` gains nothing at all (−0.021). At $k = 200$ detection falls 0.991 $\rightarrow$ 0.623, back inside the coarse-unit band that granularity was supposed to escape, while routing accuracy collapses 0.966 $\rightarrow$ 0.343 and `key_exact` routes at 0.025 on *every* name-free surface — a name matcher and nothing else. The behavioural family replicates this: `activation_norm` 0.877 $\rightarrow$ 0.495 and `attn_norm` 0.758 $\rightarrow$ 0.519, i.e. chance, on both $k = 200$ pools.

The exception. `pp1` does not collapse: 0.783/0.769 name-stripped on the rank-32 pools. All three behavioural selectors *run* the candidate experts; the two that collapse read activation *magnitudes*, and the survivor reads the expert’s *loss*. So “runs the expert” is necessary but not sufficient, and the leak is a property of the cheap selectors people actually deploy rather than something unavoidable — which is what makes a record-free defense worth pricing (§9).

A learned probe adds nothing. Fitting a logistic reader over permutation-invariant functions of the surviving scores reaches AUC 0.990 at $k = 200$ with no deletion record — and its median lift over plain confidence is +0.001. D3 fails, but it fails to a threshold, not to a learned residual trace. Under every hard transform the same probe sits at its own shuffle control (0.446 vs. 0.503; 0.417 vs. 0.501; 0.414 vs. 0.532).

Steering the selector. Because routing is lexical, it is also *attackable*. Appending a single chosen author name to a query captures 97.7% of queries against `key_exact`, 31.7% against `key_tfidf`, but only 3.5% against `centroid_sbert`; *substituting* the name captures $\sim 87\%$ in every family. Composed with §6, an adversary against a lexical selector chooses *whose* biography gets attributed to the erased person.

Pool	Transform	catch 0.50	catch 0.90	catch 0.99
r32/e25	gold	0.000	0.000	0.000
	indirect	0.019	0.125	0.236
	name_stripped	0.120	0.418	0.654
r8/e5	name_stripped	0.361	0.639	0.769

Table 4: False-refusal rate on retained traffic at a given orphan catch rate, for the `pp1` refusal gate. The gate is perfect where it is not needed and unusable where it is.

9 The Defense Frontier

We ask whether the one surviving detector, `pp1`, yields a deployable record-free refusal gate. The section is a sequence of four results and it ends negatively.

1. It can be made cheap. Scoring `pp1` over all 180 survivors is 180 forward passes per query. A free `key_tfidf` prefilter cutting to the top- m candidates costs nothing measurable: $m = 4$ ($45\times$) under `name_stripped`, $m = 2$ ($90\times$) under `indirect`. Detection does not require *correct* routing — at $m = 4$ the prefilter loses the query’s own expert 23% of the time and detection is unaffected, because orphans are uniformly poorly fit across whatever candidates are scored.

2. The prefilter improves detection rather than trading it away. Under `indirect`, AUC is monotone *decreasing* in m (0.9555 at $m = 2$ down to 0.8545 at $m = 180$). The score is a *min* over candidates, so each additional expert is another chance for an orphan to find an accidental good fit; restricting the candidate set denies orphans that lottery while retained queries keep their own expert. Cost and accuracy point the same way, which is rare enough to state plainly.

3. It is not deployable. An AUC cannot say whether a gate is usable. Table 4 gives the best achievable false-refusal rate on *retained* traffic, minimised over m .

A gold-form query *names the author being asked about*; if the deletion request was honoured at all, that name is the one thing the system is known to hold about the deleted party. The adversary who matters does not name their target — and against them the gate refuses nearly half of everyone else’s traffic. **Cost was never the binding constraint:** we solved it, $45\text{--}90\times$, and it turned out not to matter. The frontier is bounded by discrimination.

4. The apparent hard case was not hard. `indirect` replaces the name with a description built from the author’s *own* distinctive facts, which in practice prepends their book titles. It is therefore nearly as lexically self-identifying as the gold form: `key_tfidf`’s top-2 contains the true owner 0.965 of the time, against 0.998 gold and only 0.685 name-stripped. Decomposing routing accuracy on `key_tfidf`, with the name-keeping paraphrase as the control showing that rewording alone does not hurt ($0.973 \rightarrow 0.989$): removing the name costs -0.413 , additionally rewording the rest costs a further -0.090 , but `indirect` costs only -0.254 — the description hands $\sim +0.25$ back. *A transform built from the target’s own facts cannot test anonymity.* We introduce `para_stripped` (TOFU’s own paraphrase with names removed; median 12 words, 0/800 retaining any name form) as the honest independent surface, and report `indirect` beside the gold-form column rather than as the hard case.

Selector	e5	e25	e50	reads
pp1	1.000	0.999	0.996	the expert’s loss
activation_norm	0.934	0.608	0.515	magnitudes
attn_norm	0.700	0.554	0.569	magnitudes

Table 5: Gold-form detection AUC at fixed rank 32 across the epochs axis. Same code path reproduces the published e5/e25 cells exactly, so the e50 column is comparable rather than merely adjacent.

A warning that must travel with the cost result. On the rank-8 pool the cheap gate “matches full” only because full is *poor* (0.63/0.65 AUC). A 90× speedup at AUC 0.62 is not a defense.

10 Training Longer Moves the Leak; It Does Not Remove It

Behavioural detectability is strongly recipe-dependent, and the axis is training *duration*, not capacity: holding epochs at 5 and raising rank $8 \rightarrow 32$ does not degrade `activation_norm` ($0.877 \rightarrow 0.934$), while holding rank at 32 and raising epochs $5 \rightarrow 25$ collapses it ($0.934 \rightarrow 0.608$). We built a third pool to add a fixed-in-advance third point.

`activation_norm` falls monotonically to 0.515 — **chance**. Read alone this is the cheapest defense in the literature, costing nothing but epochs. The same table refutes that reading: `pp1` is untouched at 0.996. Training duration does not remove the deletion signal; it removes it from the cheap magnitude-reading selectors and leaves it fully intact for the one that reads behaviour. An operator who trains longer has made deletion undetectable *by the weak probes* and no less detectable by the strong one. **“Train longer” must not be offered as mitigation.**

This also makes `pp1` the exception on two independent axes — query transform and training duration — which is why §9 rests on `pp1` specifically rather than on “behavioural selectors” as a class. One counter-current: name-stripped `pp1` declines gently along the same axis ($0.783 \rightarrow 0.769 \rightarrow 0.737$), so the exception is far more robust, not immune.

11 Measurement Constraints

Each constraint below invalidated a reading we had already written down. We state them because they generalise to anyone auditing a routed system.

1. **A paired quantity needs a paired interval.** The destination arms score identical rows (inter-arm $r = 0.88\text{--}0.94$). Bootstrapping each arm’s marginal re-adds their *shared* question noise once per arm and declares any spread unresolvable. Our own first verdict on §5 made exactly this error, and it flipped on the paired test.
2. **Quote the exact resolution, not a sampled one.** The “achievable p -values” of a two-sample KS test, enumerated by random draw, grow with the draw count ($73 \rightarrow 88$ between 2k and 60k draws, unconverged) — so such a count states nothing. Every attainable D is a multiple of $1/\text{lcm}(n, m)$; use that.
3. **Structural holes in the corpus concentrate in the conditions you most want to report.** The 18 name-free authors distorted three separate results, and they are the same units that act as routing magnets. Flag or exclude them in any per-survivor statistic.

4. **A transform built from the target’s own facts cannot test anonymity (§9).**
5. **A sub-chance AUC means nothing without its own shuffle control.** Two apparent systematic sign flips were noise; the control spans 0.336–0.532 here, so differences below ~ 0.1 are not resolvable and must not be quoted as values.
6. **A control over a variable the measurement does not consume is not a control.** Our first recipe ablation was vacuous by construction: feature-space selectors read no expert weights, so three pools returned byte-identical matrices.

Defects that produced plausible numbers. Eight, all recorded in the campaign log rather than quietly fixed, because every one of them yielded output that looked right: a lazy adapter cache that silently scored non-resident experts as exactly 0.0; a route audit whose assertion ran *before* the artifact was written, destroying the arm it audited; a cache race that let a sibling arm read a 0-byte file; a consolidator that paired results with the wrong generation dump; an entire query condition made unreproducible by `sorted(set, key=len)` under hash randomisation; and the two statistical errors above. The countermeasure that kept working was computing the same quantity a second way.

12 Related Work

Exact unlearning by construction. SISA [Bourtoule et al., 2021] converts deletion into bounded retraining by sharding; LegoNet [Yu et al., 2022], S3T [Basu Roy Chowdhury et al., 2025], FedSGT [Zhang et al., 2025] and continual-learning forgetting services [Huang et al., 2025] carry the recipe — isolation plus bounded recomputation — into adapters, PEFT slices, federated rounds and analytic classifiers. Merging-based variants [Ilharco et al., 2023, Yu et al., 2024, Kuo et al., 2025] make deletion an exact subtraction, and a parallel line moves the adapted knowledge out of the weights entirely, into context that can be edited [Pawelczyk et al., 2024] or removed exactly [Muresanu et al., 2024]. All of these guarantee D1. None of them, to our knowledge, reports what the composition does with the queries that used to be served by the removed component — and the in-context variants inherit the same open question, since a retriever is a selector.

Routing and serving many experts. RAMoLE [Zhao et al., 2024] retrieves and composes uploaded LoRA experts; Compress-then-Serve [Brüel Gabrielsson et al., 2025] makes thousands of adapters servable; SEA [Schneider et al., 2026] makes deletion an artifact drop; memory adapters [Grimes et al., 2026] make it a block-list entry; SEUF [Zhuang et al., 2024] asks whether editing one MoE expert suffices, and finds the gate itself is the obstacle. Our results say the selector is not only an obstacle to unlearning *inside* a module but the component that determines what a deleted source’s queries are answered *with*.

Localization. Maini et al. [2023] show memorization is localizable but not layer-confined; gradient routing [Cloud et al., 2024], MemSinks [Ghosal et al., 2025] and capability-removal-by-localization [Shilov et al., 2025] engineer localization so that a component can be removed. This paper is orthogonal and complementary: it grants perfect localization and asks what the surrounding system does afterwards.

Benchmarks and attacks. TOFU [Maini et al., 2024] is the yardstick, and §5 is a negative result about it in the routed setting specifically. Wu et al. [2025] show that exact unlearning of the final model does not prevent extraction from the broader pipeline; our failure is complementary and requires no pipeline access at all — the deployed system emits it in the normal course of answering.

13 Limitations and What We Do Not Claim

- **One benchmark, one base model, one deletion size.** TOFU’s fictitious authors give clean ground truth about who knows what, which is exactly why the fact-level CSAR metric is computable at all; they are also uniform, synthetic and English. We deliberately do not claim these rates transfer to a heterogeneous production corpus.
- **We do not claim any destination is better or worse than another.** The per-destination ordering does not reproduce across evaluation tiers, only the spread does.
- **CSAR is not yet validated. [BLOCKED: ~300 hand labels outstanding.]** The decomposition into substantive and name-only inherits the classifier’s errors, and because 18 authors yield no extractable name, “substantive” is an upper and “name-only” a lower bound.
- **We do not claim no record-free defense is possible.** §9 bounds ppl-as-refusal-gate at $k = 200$ on three pools under one transform family.
- **We do not claim training duration mitigates anything** — §10 is a warning against precisely that reading.
- **The privacy/MIA column is withheld.** In one wave every arm reported byte-identical attack AUCs while serving different models, which we attribute to all queries falling to the out-of-distribution path; rather than publish a number we cannot explain, we report its absence. [TODO: re-run the MIA battery with route stats recorded per arm]

14 Conclusion

Modular unlearning delivers what it promises: after deletion, no surviving parameter was trained on the removed source. It does not deliver what the word “deleted” is heard to mean. The queries survive the module, the selector reassigns them, and the system answers them with a stranger’s life — specifically, fluently, and without refusing. TOFU’s forget metric scores that outcome at or above genuine deletion, and the properties that make the pattern look safe — detectability, locality, refusability — are all measured on queries that name the person being deleted, and all degrade when they do not.

We therefore propose that a routed system’s deletion claim is incomplete without a substitution report: what fraction of orphaned queries are answered (ORR), with whose facts (CSAR), and at what collateral cost to retained traffic (RDR). Deleting the module is the easy half. Deciding what the system says next is the deletion problem that modular architectures have quietly left open.

Reproducibility. Every number in this paper is produced by a CPU-gated analysis over score matrices and generations that are committed alongside the code; [TODO: insert the anonymized artifact URL at submission time].

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